



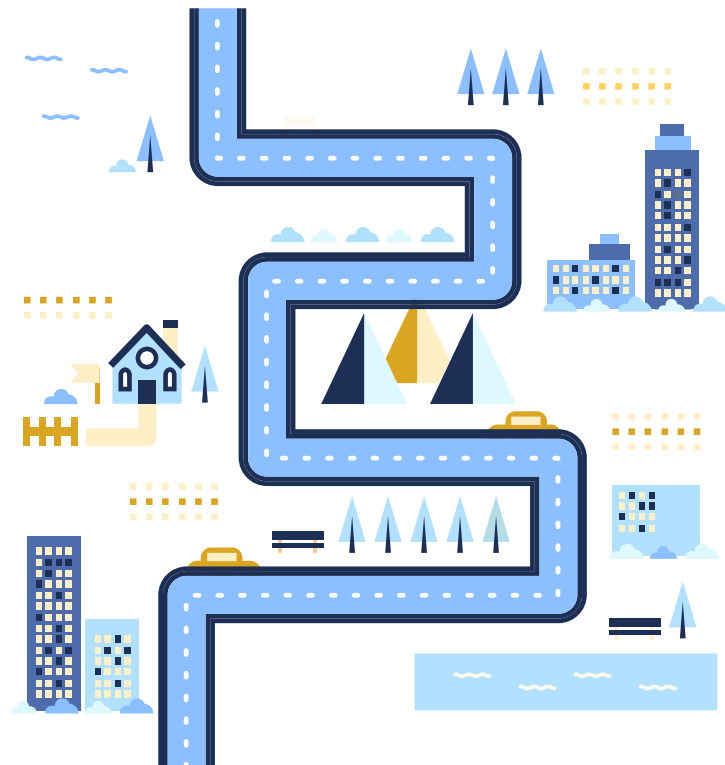
Optimizing Transit Expansion with Network Metrics in Chicago

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Introduction: The Problem

Problem:

- Many modern cities lack efficient public transit
- Strategy is needed when determining where and how to expand public transportation
- Network analysis can help



Introduction: The Dataset

net.tntp

Road network nodes and links (capacity, travel time).

node.tntp

Contains the geographic coordinates of nodes in the network in Illinois State Plane coordinates



flow.tntp

Traffic flow on network links (vehicles per time unit).

trips.tntp

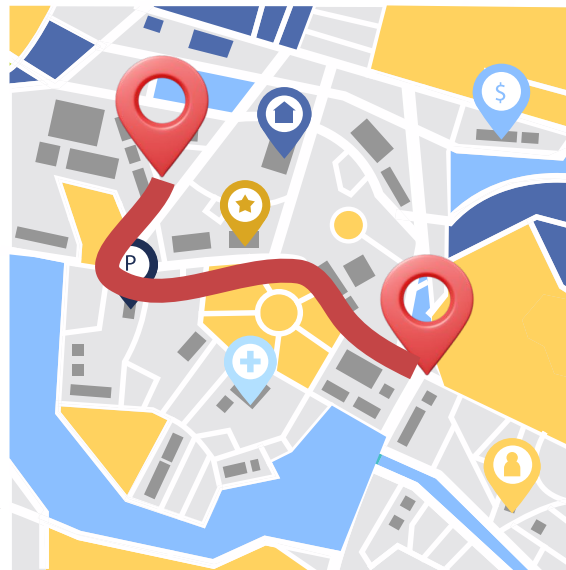
Travel demand between zones

Method 1: Base Network Analysis

- Focus solely on the structure
- Original dataset had “start” and “end” nodes

$$A_{i,j} = \begin{cases} 1 & \text{if there exists an edge from } j \text{ to } i \\ 0 & \text{otherwise} \end{cases}$$

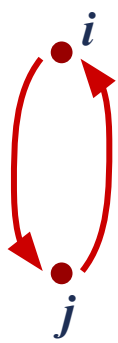
- 1 strongly connected component
- 1 weakly connected component



Method 1: Base Network Analysis

- The matrix is symmetric!

→ we can treat it as undirected



before

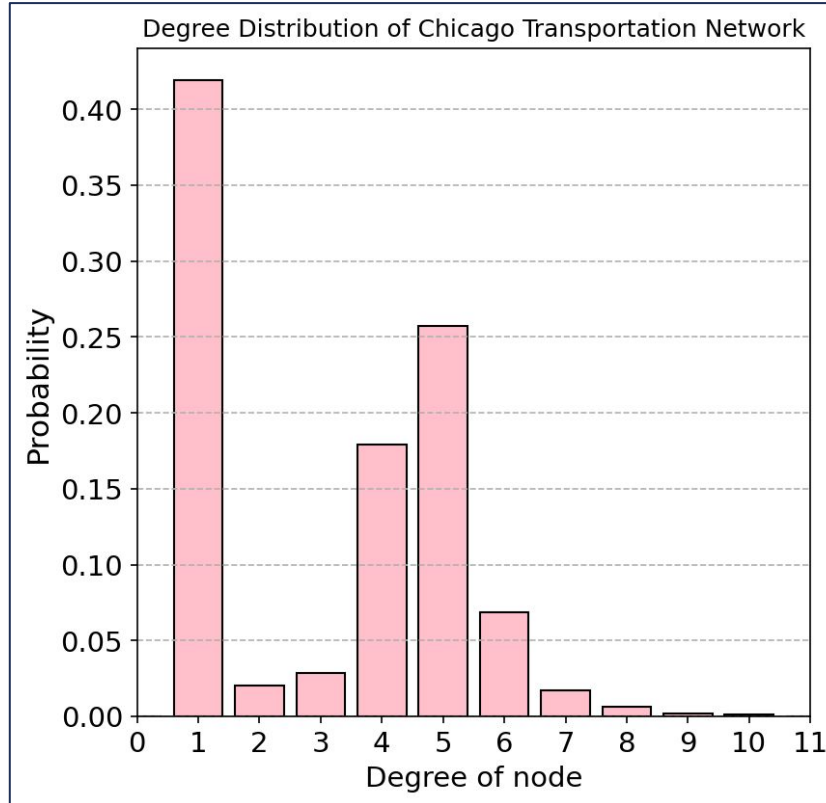


after

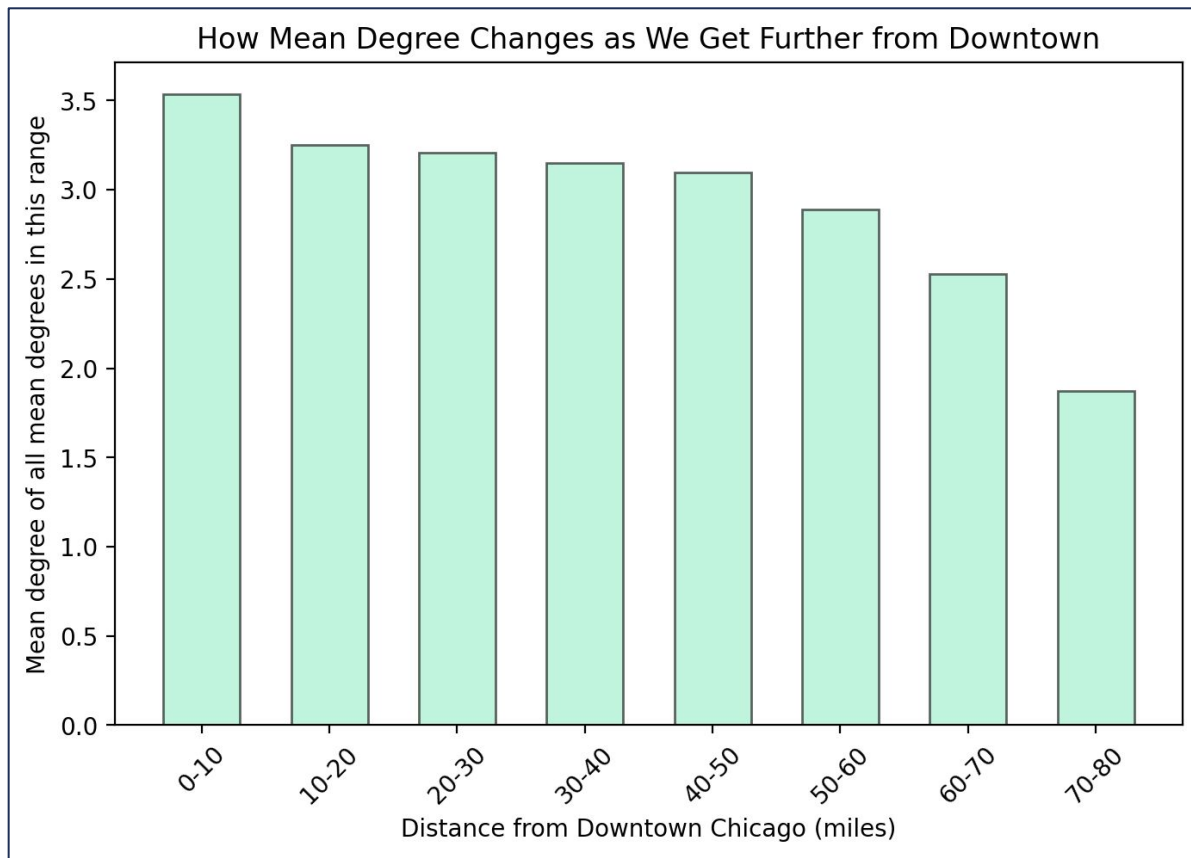
$$A_{i,j} = \begin{cases} 1 & \text{if there is a bidirectional road from } i \text{ to } j \\ 0 & \text{otherwise} \end{cases}$$

Method 1: Base Network Analysis

- Density: 0.003393
- Mean degree: 3.16



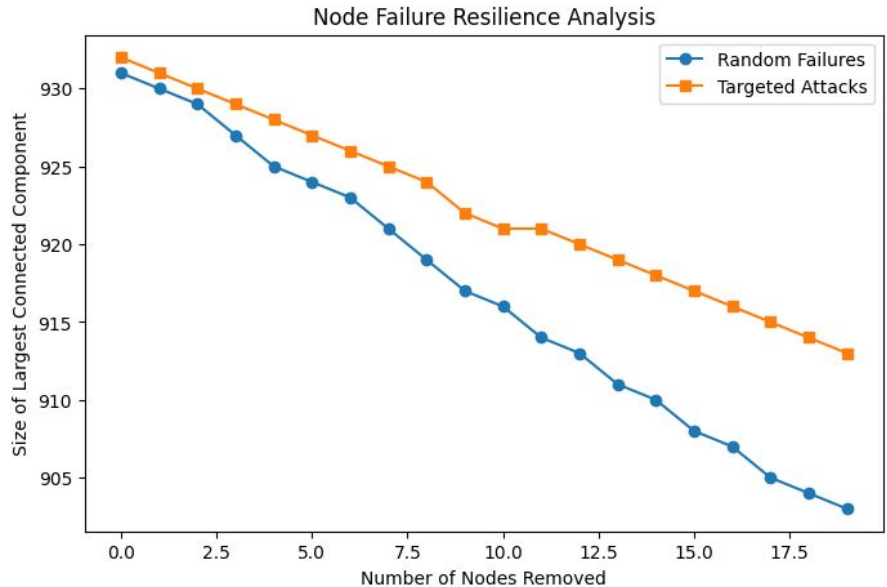
Method 1: Base Network Analysis



Closer to downtown
→ higher avg. road
connections

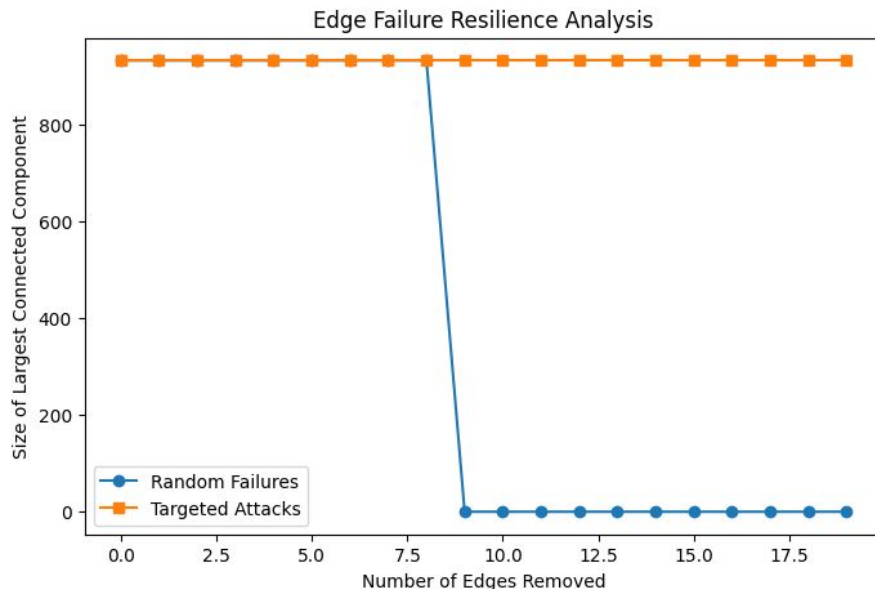
Method 2: Network Resilience Analysis

- Random failures cause a steady decline, so the network is well-distributed with no single point of failure.
- Targeted attacks degrade the network more efficiently, so certain roads or intersections are critical to overall connectivity.
- The difference between the curves shows that the network is not highly robust to strategic failures.



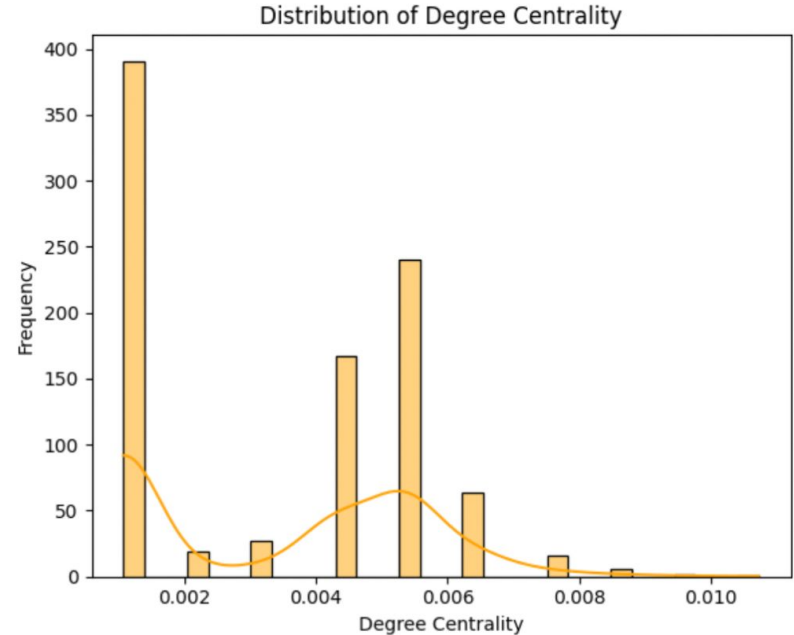
Method 2: Network Resilience Analysis

- In edge failures, the network seems more resistant to even targeted removals.
- the network heavily depends on key intersections (node-based vulnerability) rather than individual roads (edge-based strength).
- Even if high-capacity roads are removed, alternative routes keep the network functional, but if certain critical nodes fail, the whole system can collapse



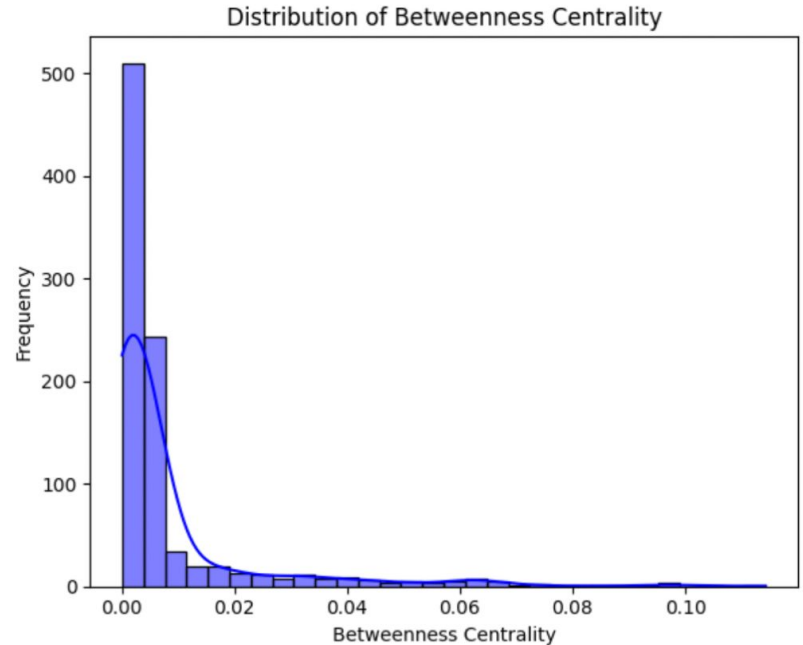
Method 3: Centrality Analysis

- Degree centrality measures direct connections
 - Most nodes have low degree centrality, but there are a few peaks with higher values
 - Representing central intersections



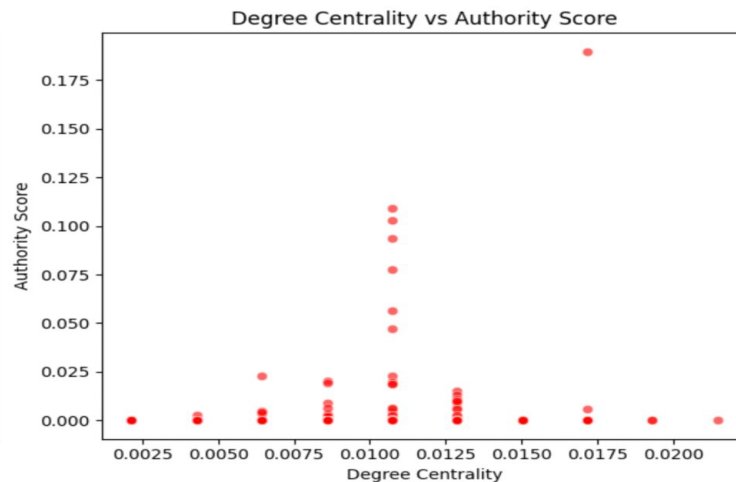
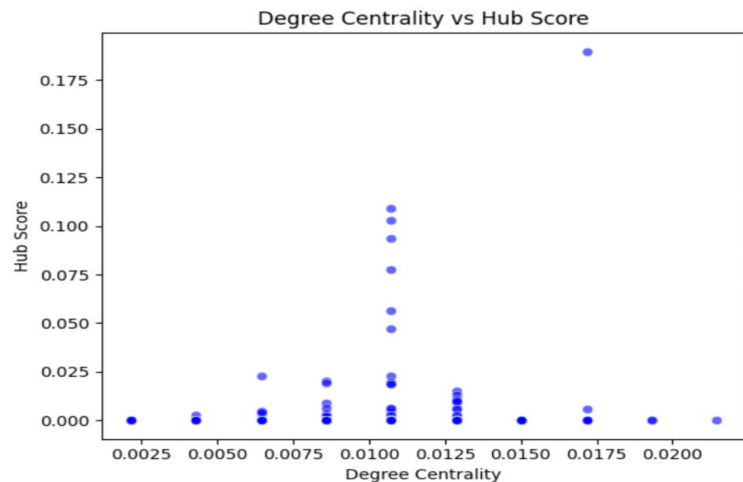
Method 3: Centrality Analysis

- Betweenness centrality measures how frequently a node lies on the shortest path between other nodes
 - Highly right skewed, with few high values
 - Indicates that there are a few nodes that play a more crucial role in the flow of the network
 - Corresponds with downtown vs. peripheral suburbs



Method 3: Centrality Analysis

- Consider hubs and authorities, want to analyze nodes pointed at and nodes pointing to other important
- Nodes have identical hub and authority scores
- Symmetric network structure leads to lack of distinction between these values



Method 4- OD Analysis

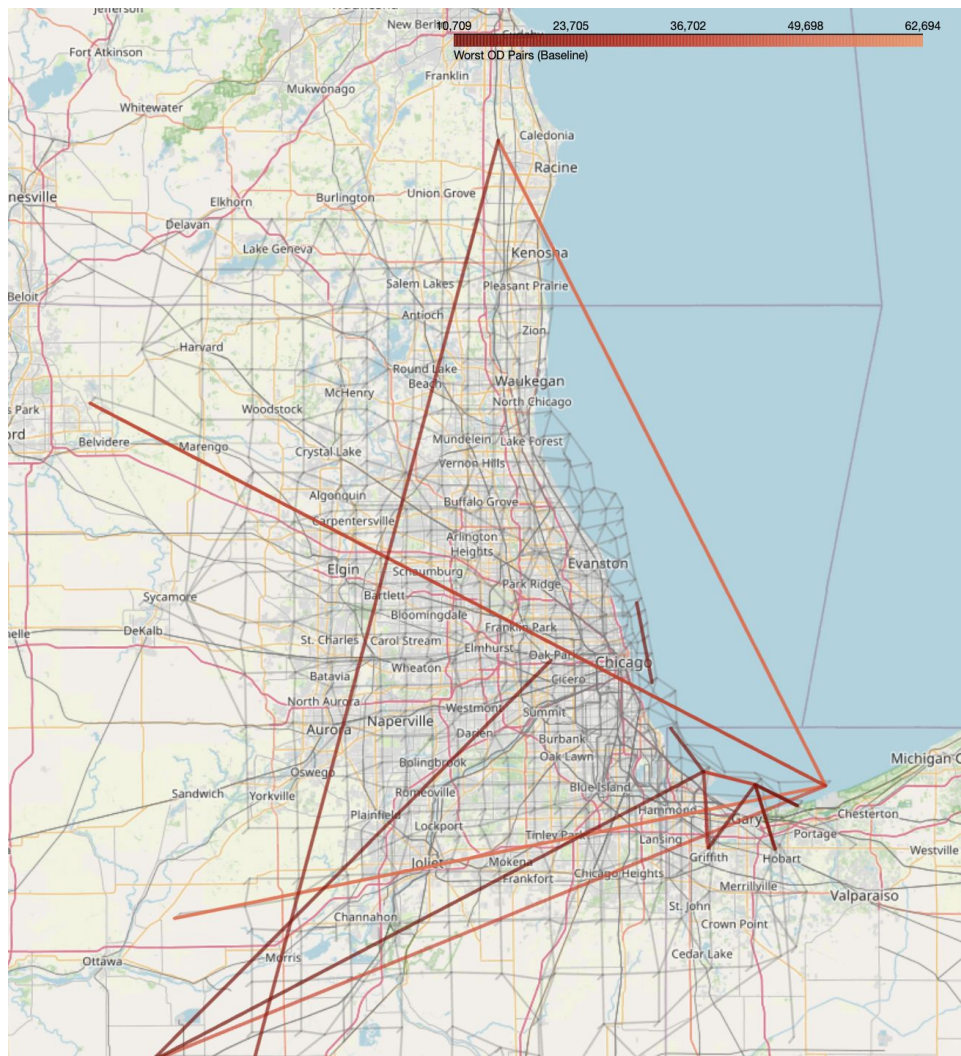
- The Origin-Destination (OD) dataset provides trip demand between different zones in the network.
- Zones $\neq$ Nodes $\rightarrow$ Zones represent aggregated areas, grouping multiple nodes based on regional demand patterns.
- We define cost of each pair as as:

$$\text{cost} = \text{demand}_{O,D} \times \text{shortest path}_{O,D}$$

- **Intuition:**
 - OD pairs with extremely high cost highlight high-demand corridors that lack efficient connectivity.
 - These bottleneck routes create significant travel delays and congestion.
- **Why does this matter?**
 - High-cost OD pairs are prime candidates for public transit expansion.
 - Addressing these corridors reduces system-wide inefficiencies and improves accessibility.

Method 4- OD Analysis

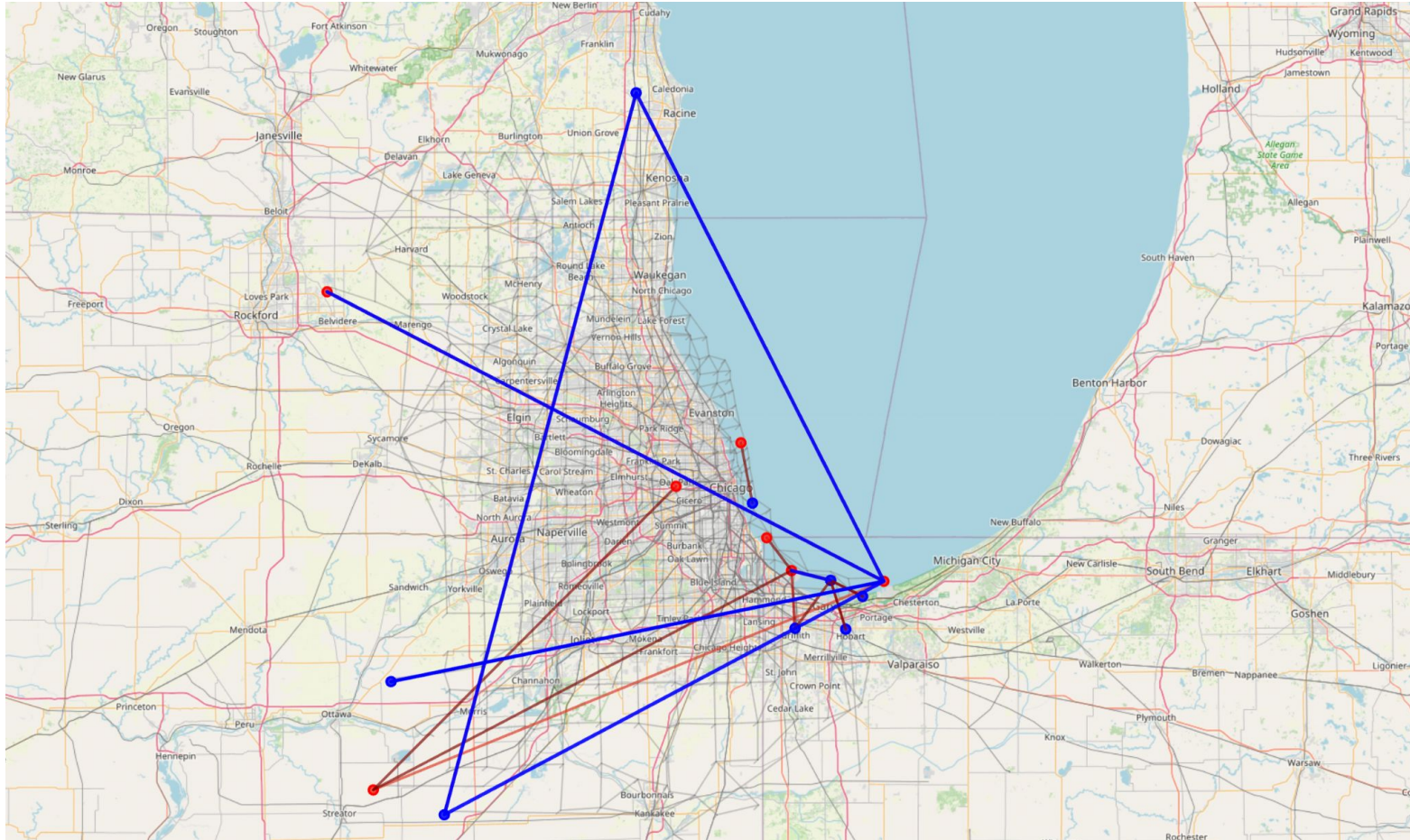
- Let's compute cost between every OD pair and find which ones are the highest:
 - The zone that corresponded to downtown appeared the most frequently in these pairs
 - This suggests node 387 is a critical hub in the network, experiencing frequent travel demand despite having high travel costs.
 - We can then plot the “worst” (highest cost) OD pairs the network, overlaid on a map of Chicago



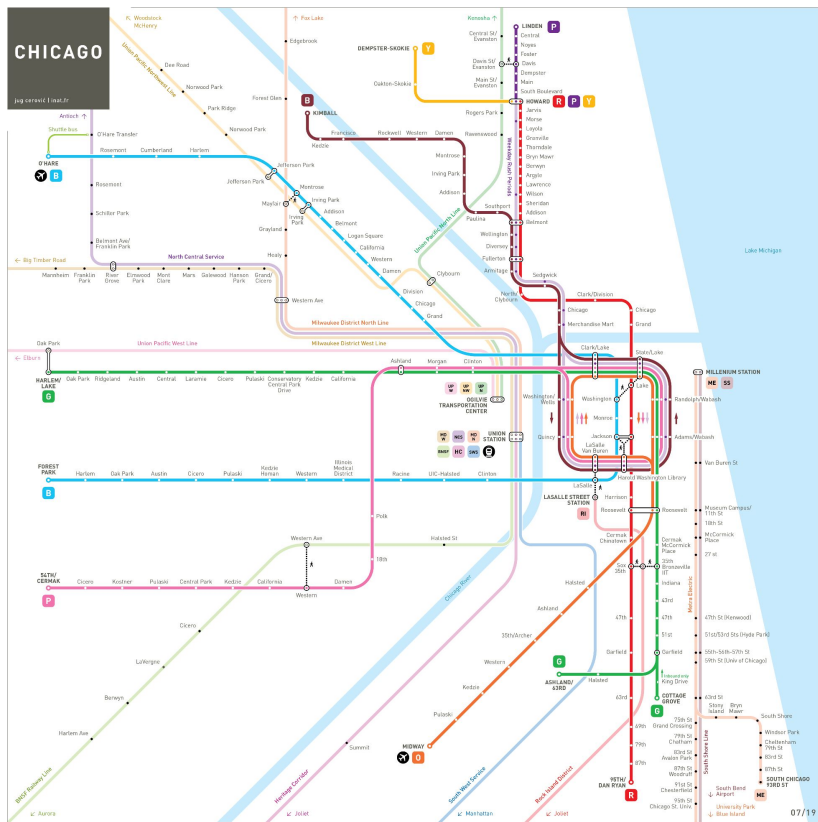
Simulate New Links

- With this insight, we simulate network improvements by adding links to the top six highest-cost OD pairs
- Simulated links serve as a simplified representation of a metro system, providing direct, high-speed connections between high-demand OD pairs
- This represents an **idealized transit network**
 - Not realistic due to simplified assumptions
- After adding these simulated links, the baseline metric we use to measure our network- average shortest path length- decreases
 - New network is better than the original

Result



Chicago Metro System Comparison



Conclusions and Discussion

- Network analysis provides actionable insights for public transit planning
- Downtown Chicago is a major hub, but suburb-to-suburb travel is also critical
- OD analysis identifies high-cost corridors where transit expansion would be most impactful
- Simple network measures can generalize to other cities, forming a scalable framework for transit planning.

